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Alvarado Valverde

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(54) **LIFT GENERATING DEVICE**

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(51) **Int. Cl.**
B64C 39/06 (2006.01)

(52) **U.S. Cl.**
CPC ***B64C 39/064*** (2013.01)

(58) **Field of Classification Search**
CPC B64C 39/064
USPC ..... 244/12.1–12.3, 23 B, 23 C, 7 R, 10, 21, 244/52; 446/36, 37
See application file for complete search history.

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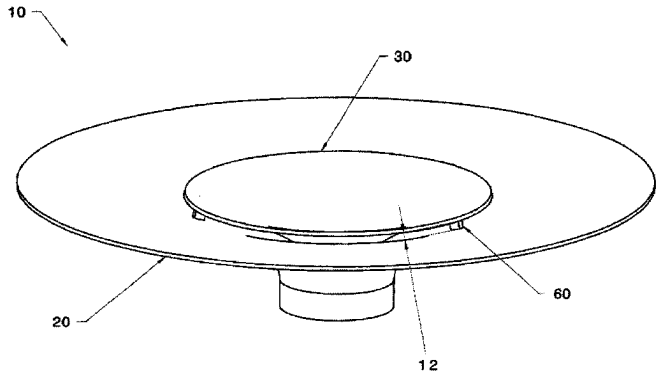
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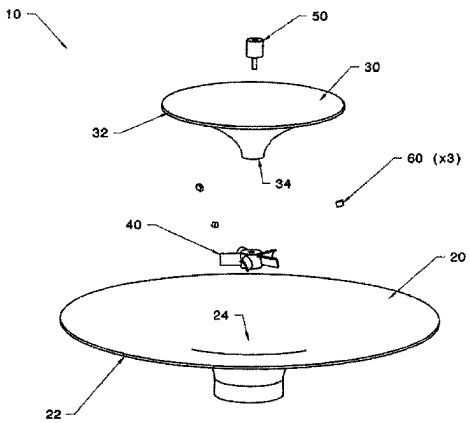
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(57) **ABSTRACT**

A lift generating device is provided enabling vertical flight. The lift generating device includes a first revolution structure having a funnel shape and a second revolution structure having a funnel shape. Each revolution structure includes a circular open surface and an open stem. The second revolution structure is attached to the first revolution structure by a plurality of connectors so as to form a gap between the first revolution structure and the second revolution structure. A propeller is coupled to an open stem at a bottom end of the first revolution structure and a motor is operatively connected to the propeller to provide power for the propeller to suction air into the lift generating device.

8 Claims, 15 Drawing Sheets





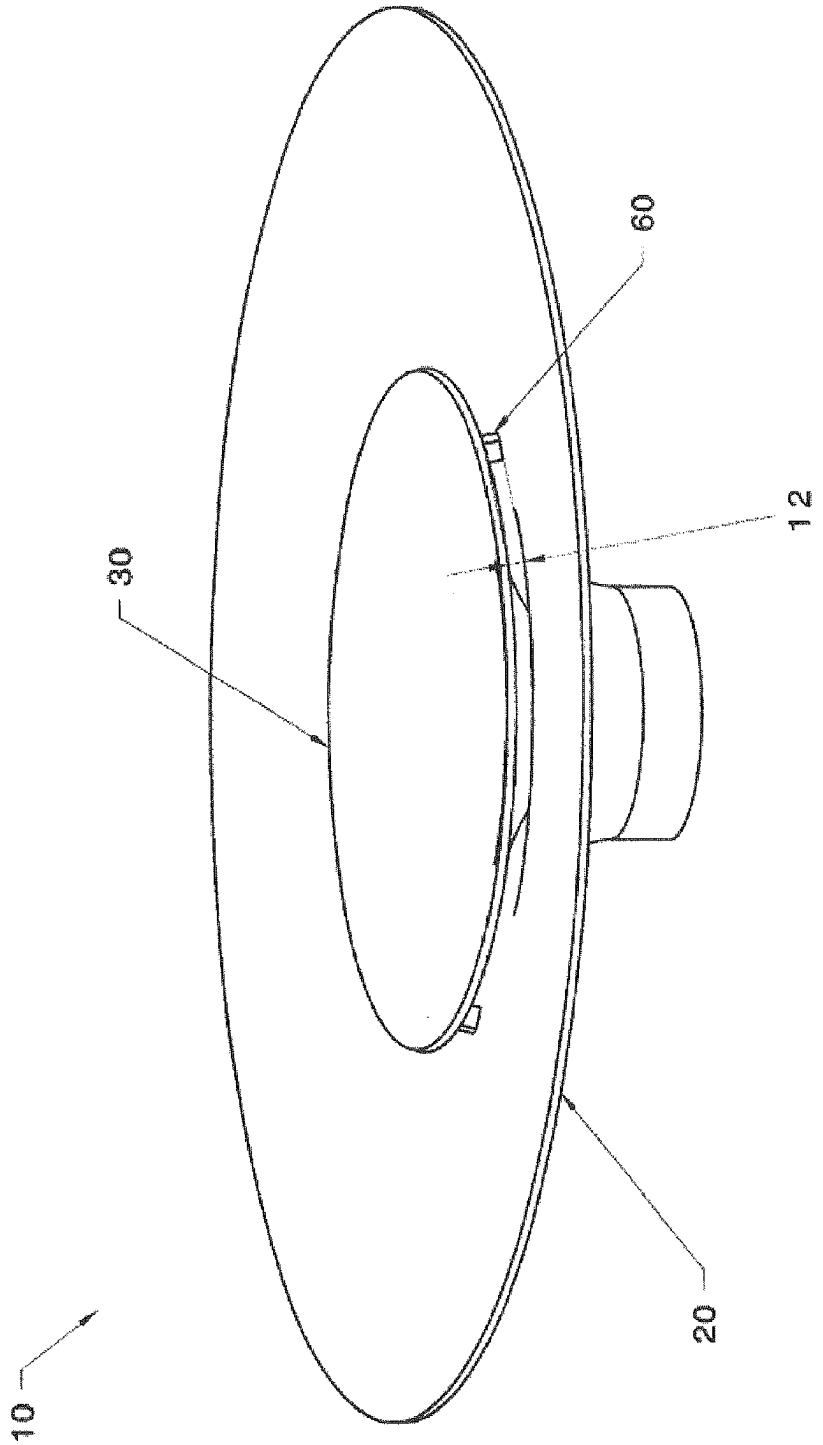


FIG. 1

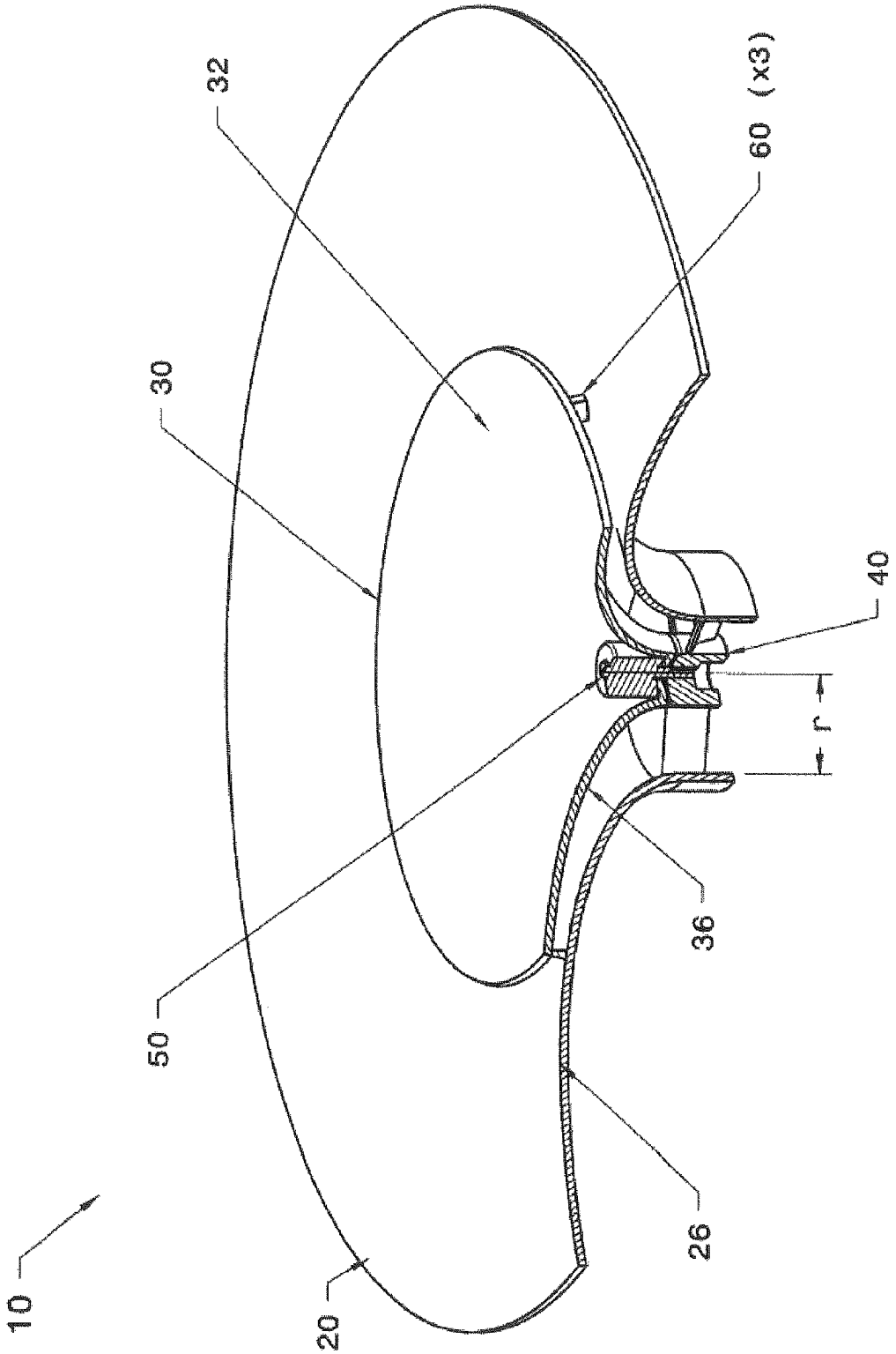


FIG. 2

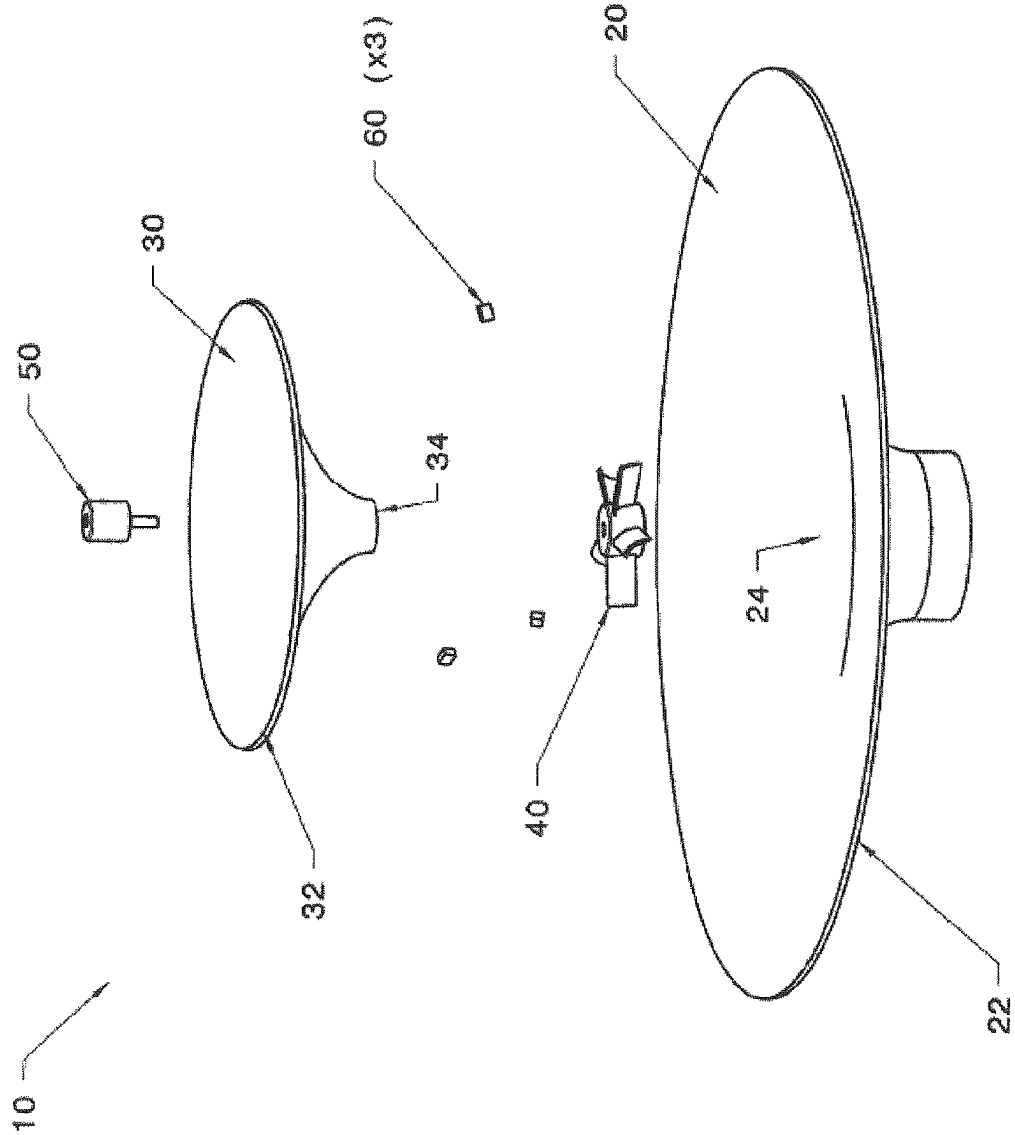


FIG. 3

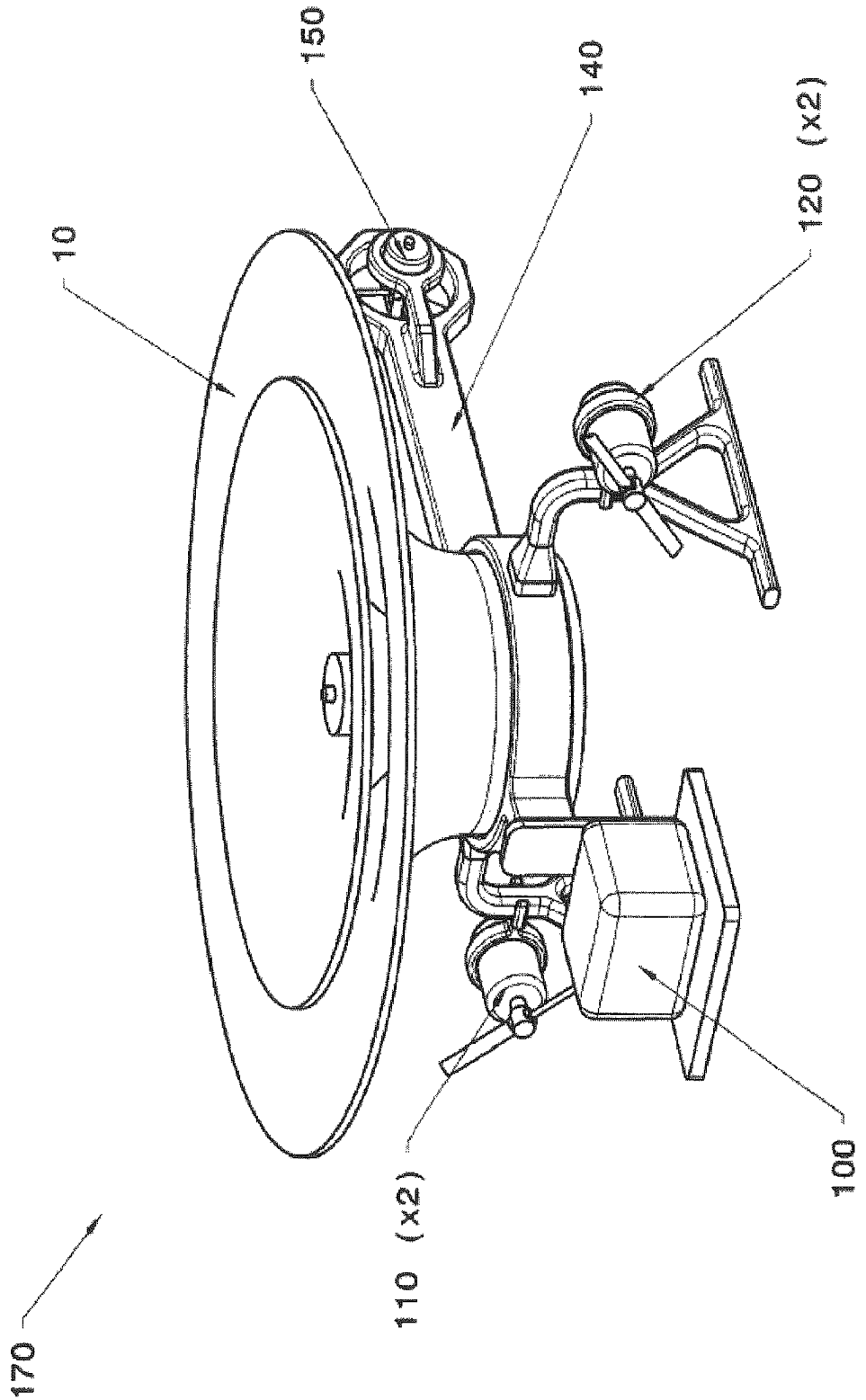


FIG. 4

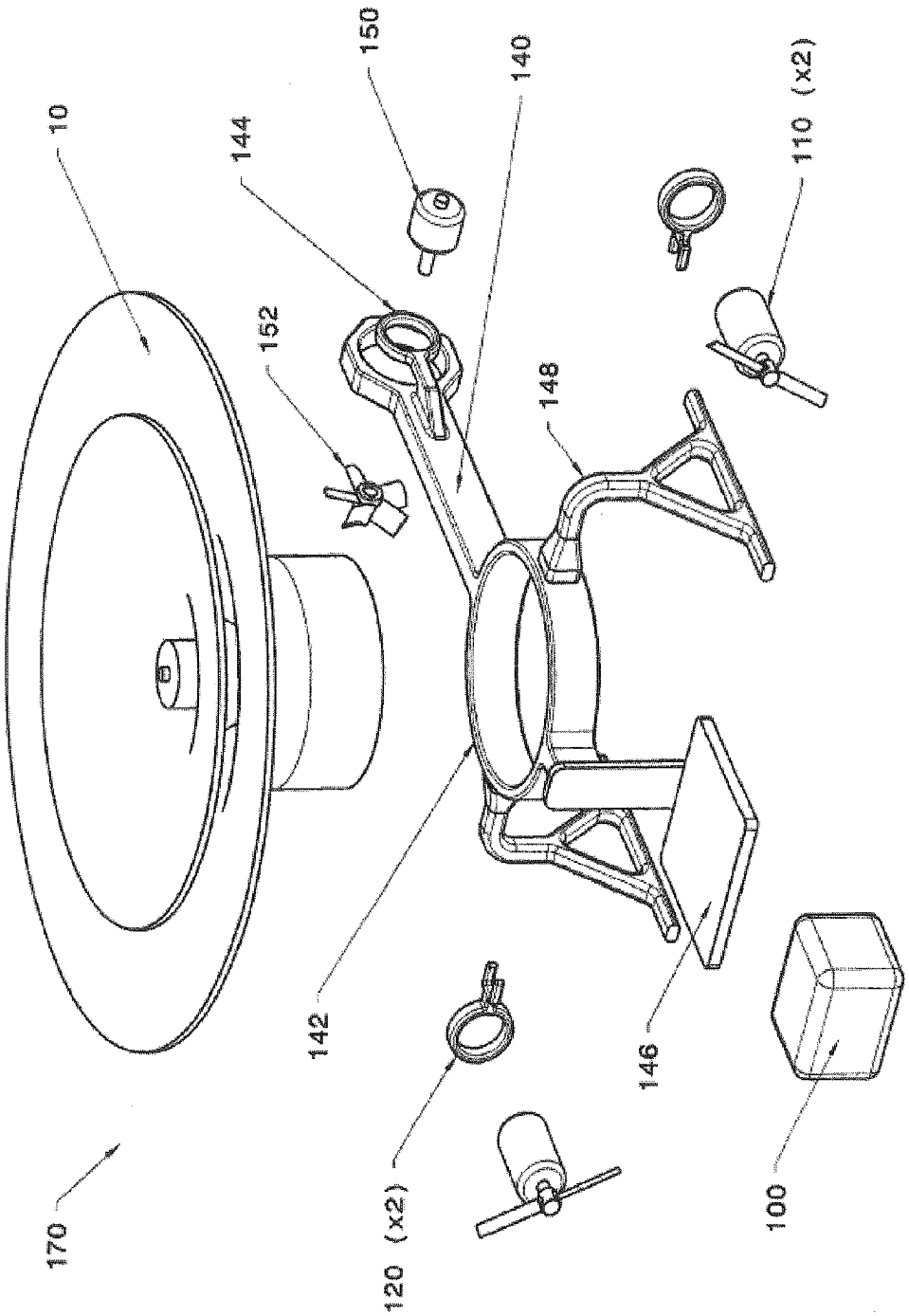


FIG. 5

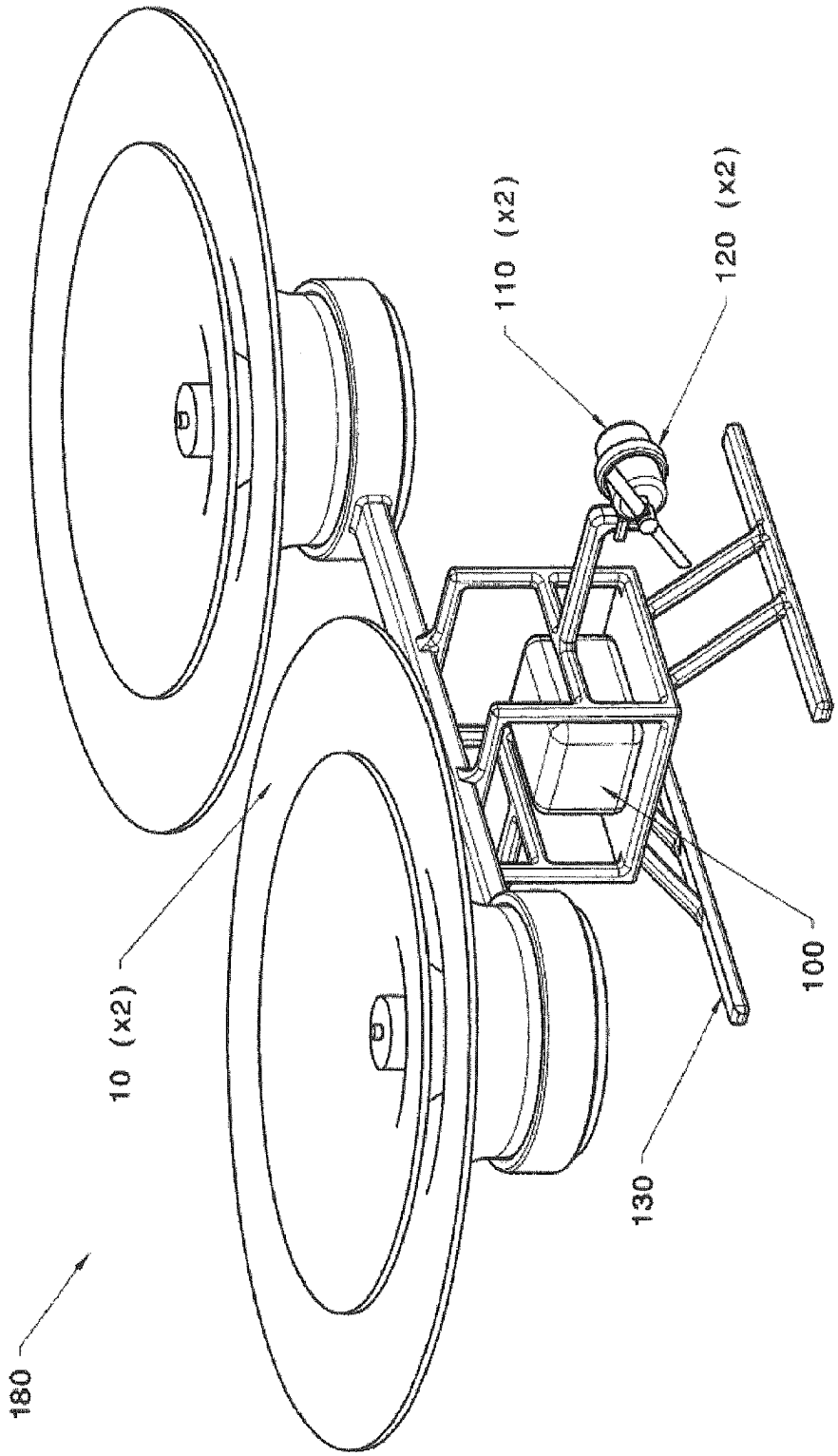


FIG. 6

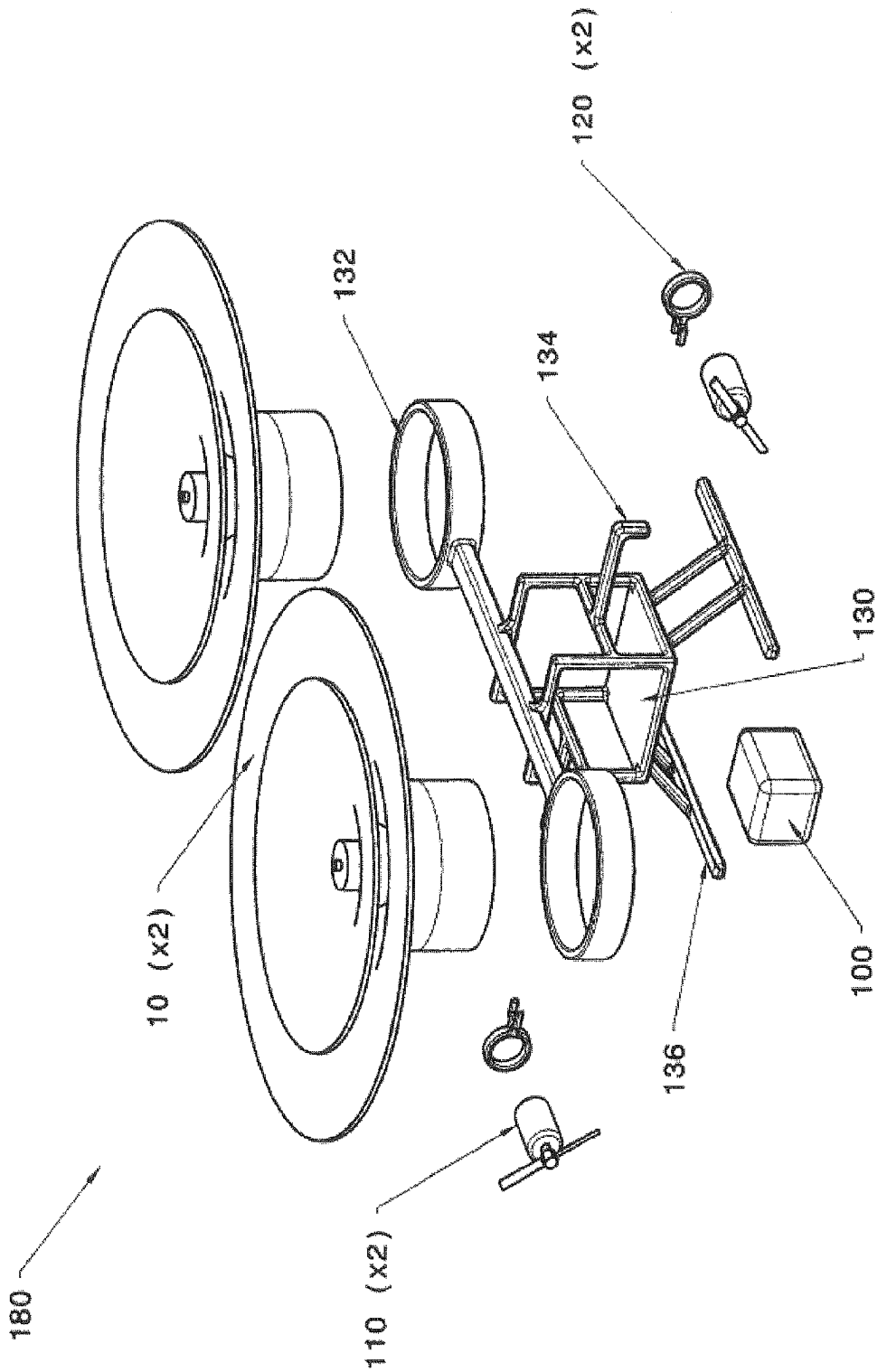


FIG. 7

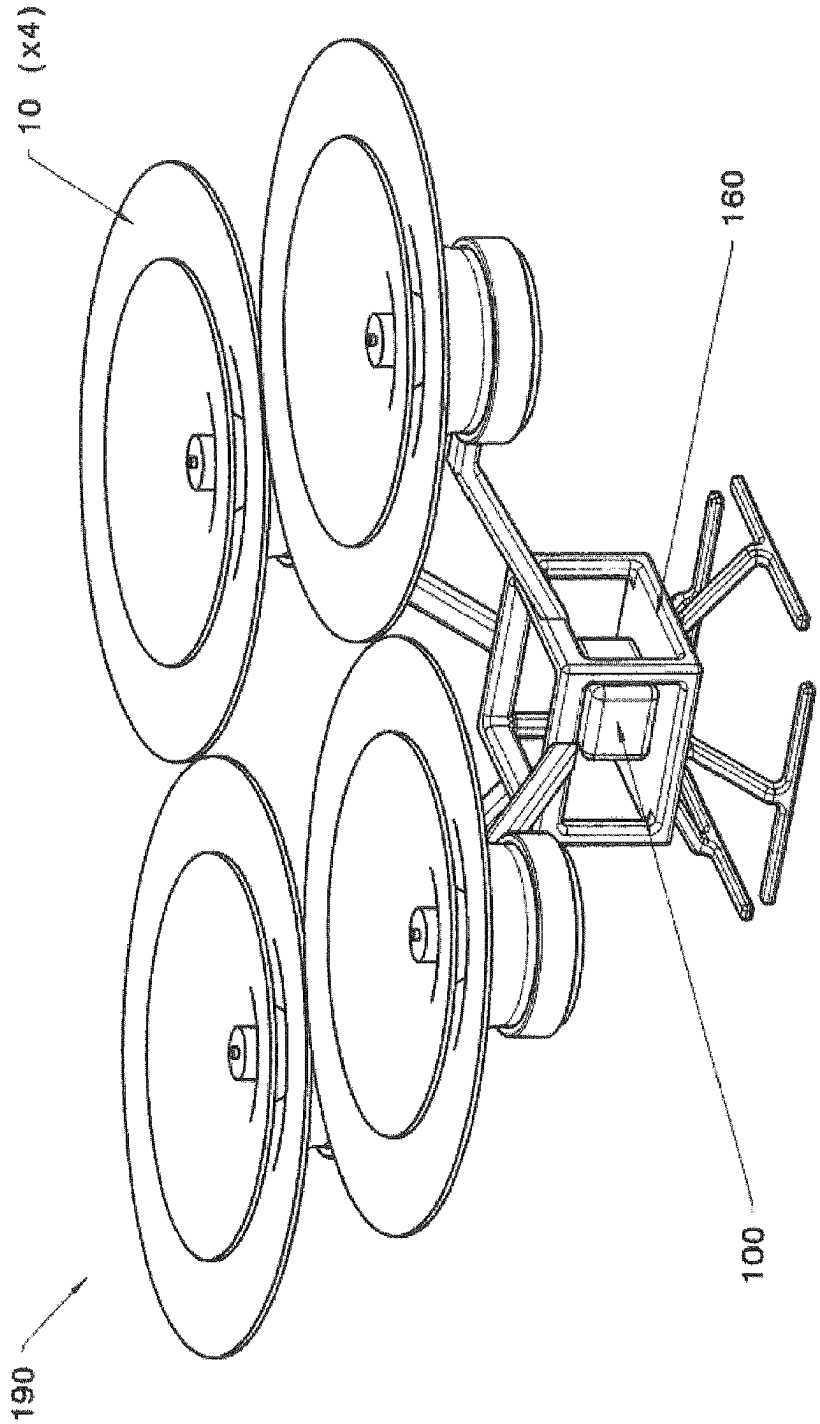


FIG. 8

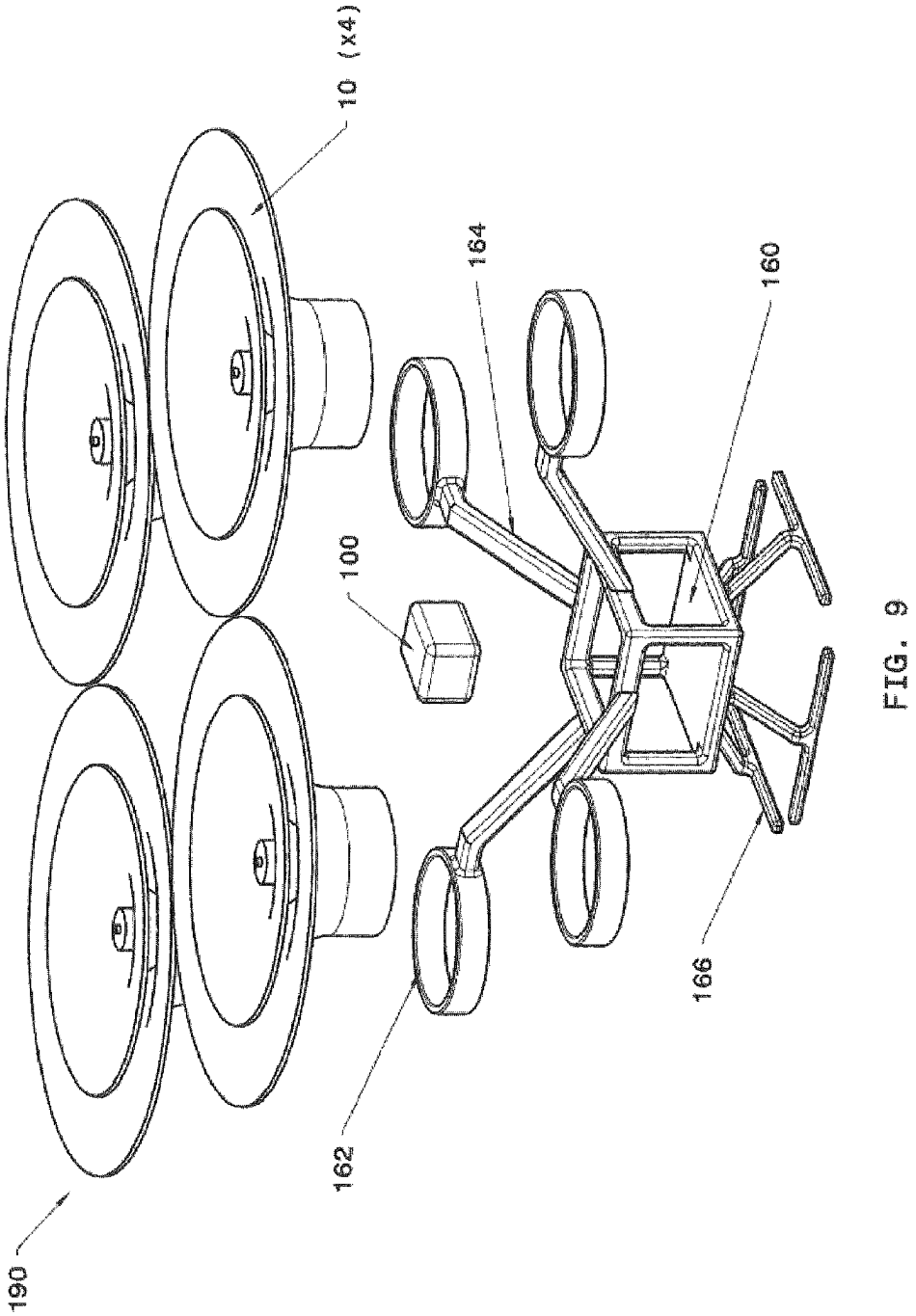


FIG. 9

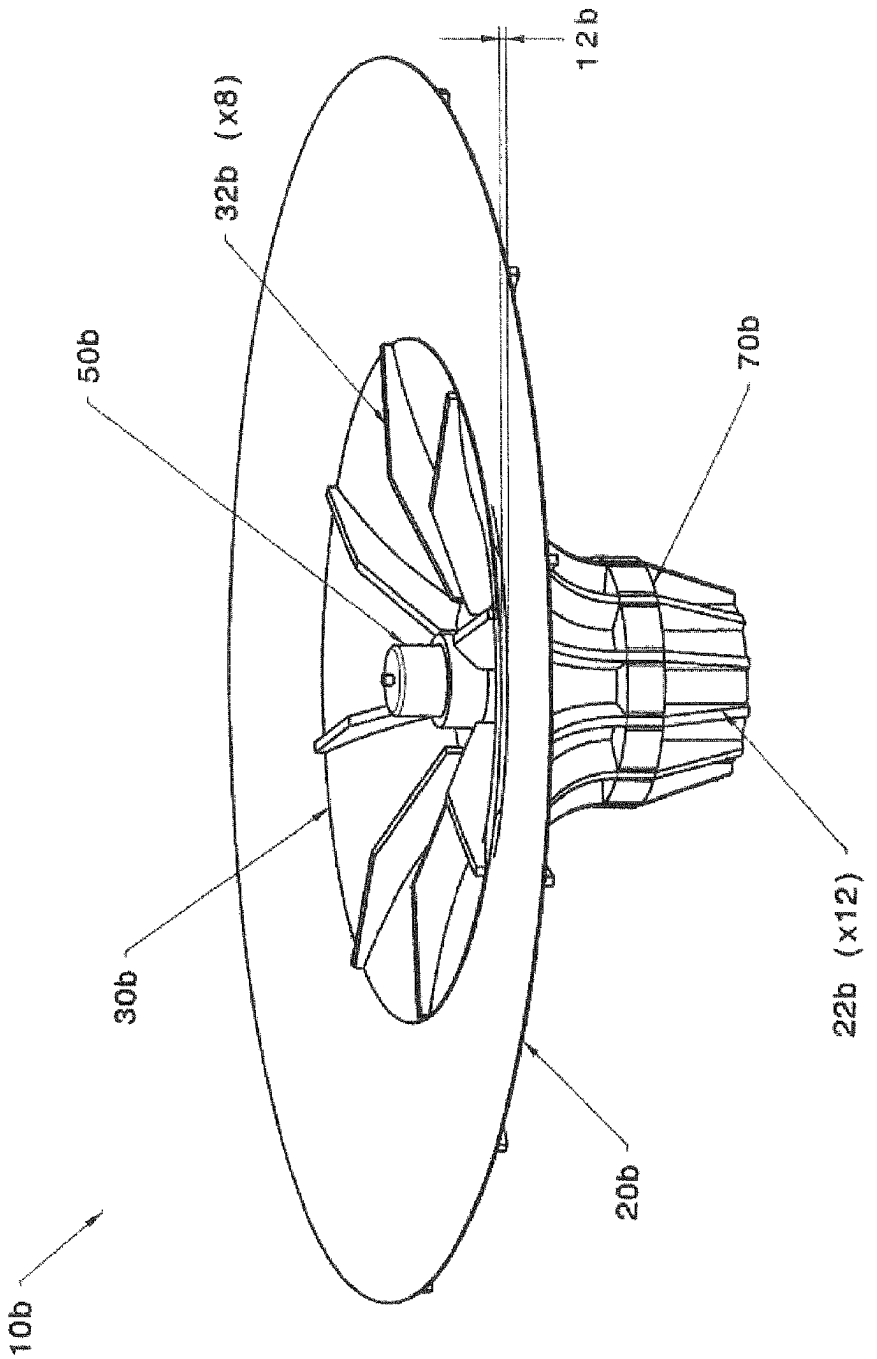


FIG. 10

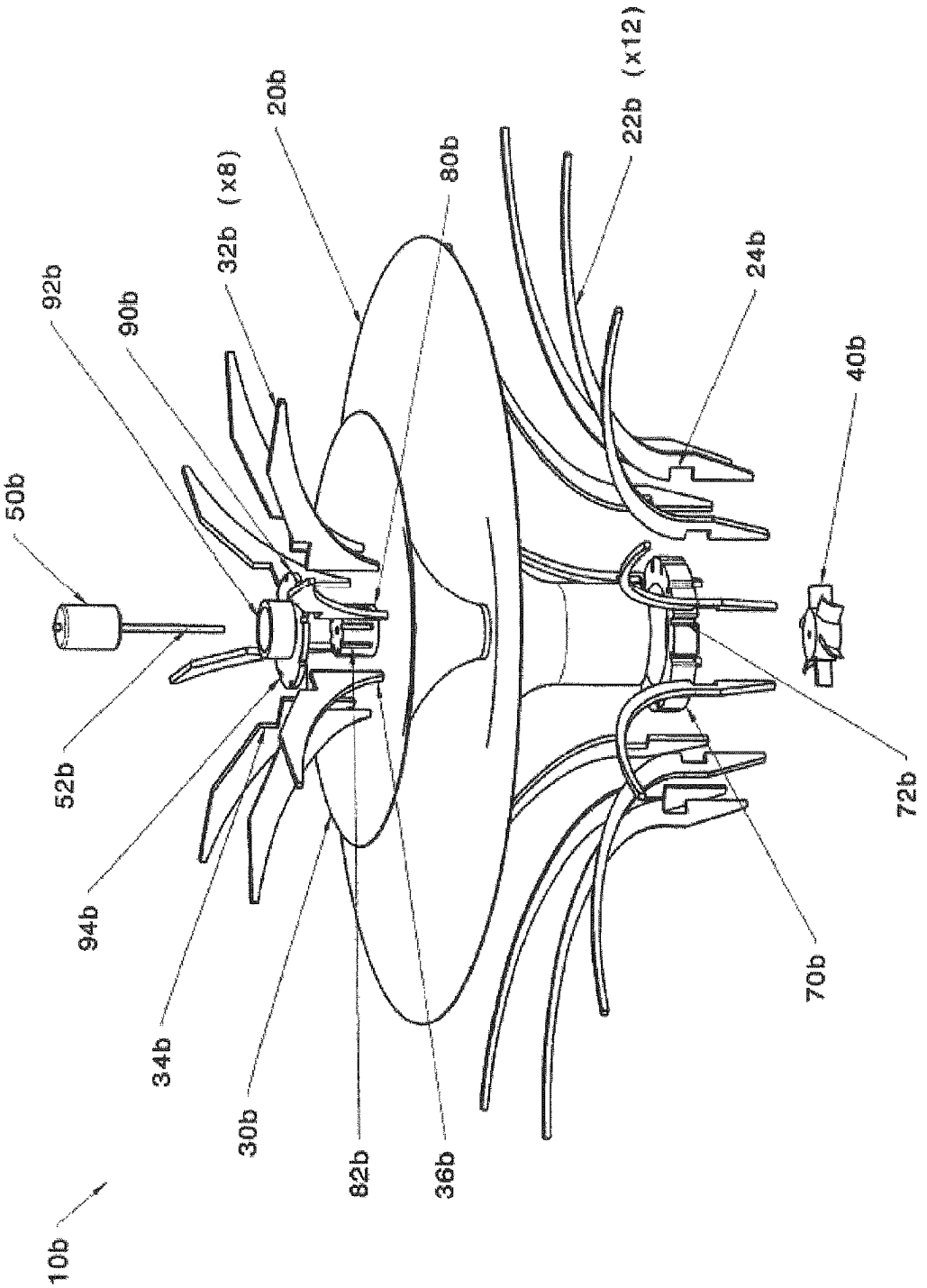

10b
20b
22b (x12)
24b
30b
32b (x8)
34b
36b
40b
50b
52b
70b
72b
80b
82b
90b
92b
94b

FIG. 11

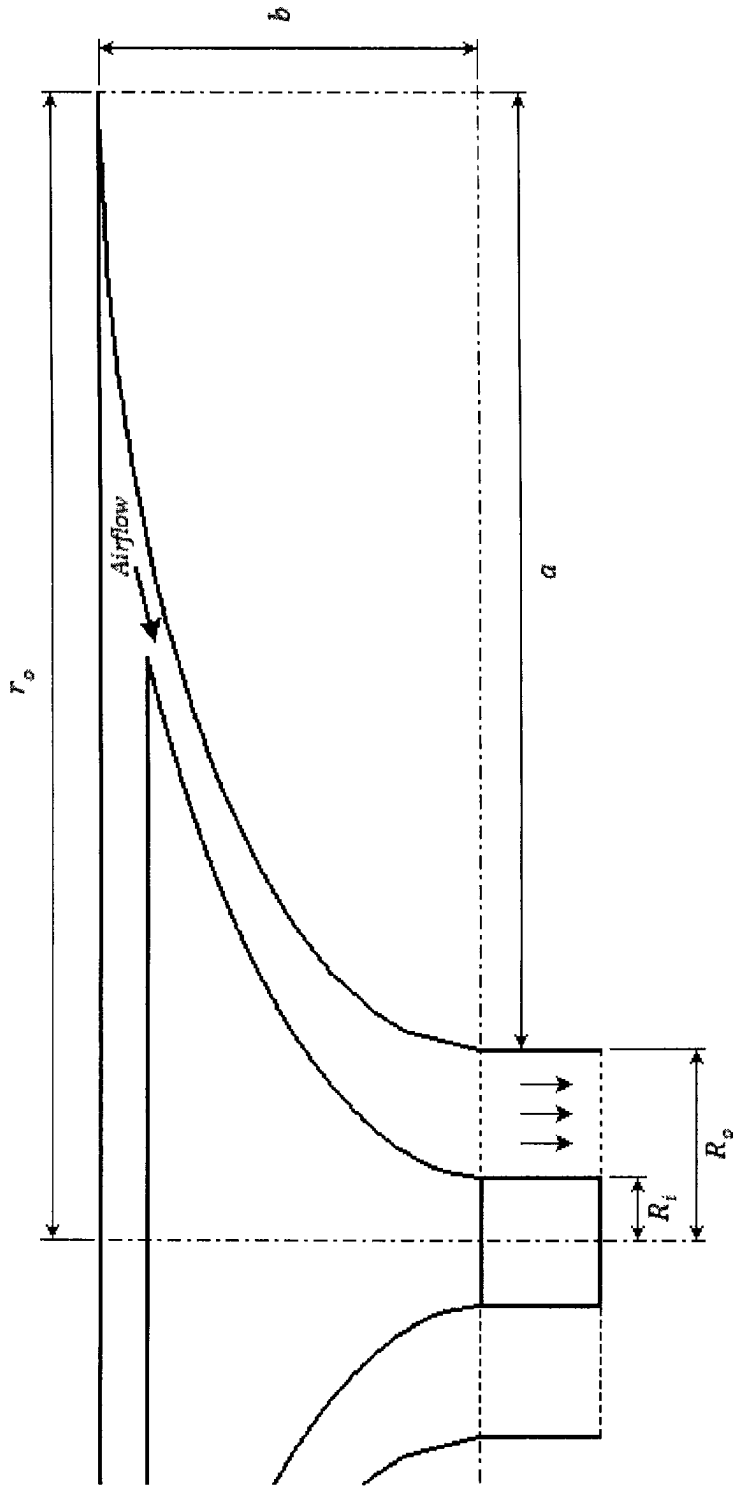


FIG. 12

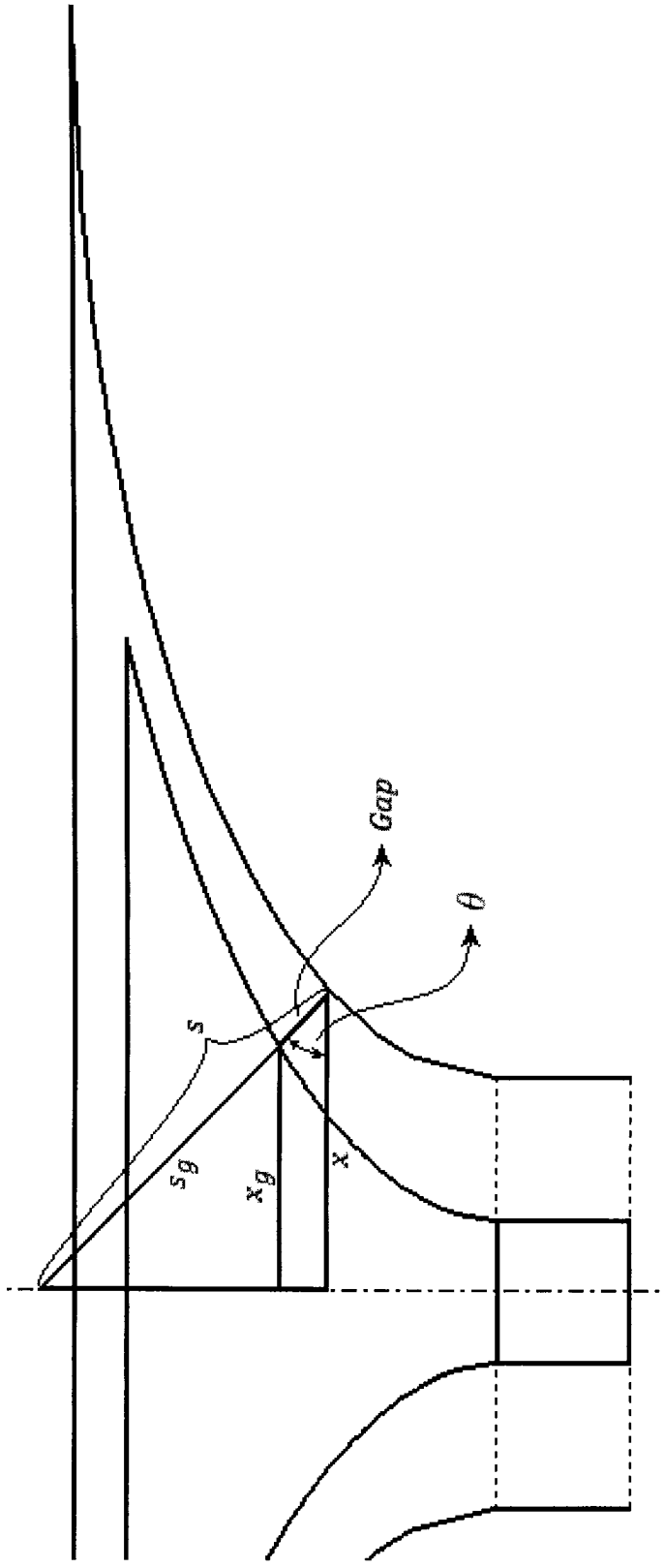


FIG. 13

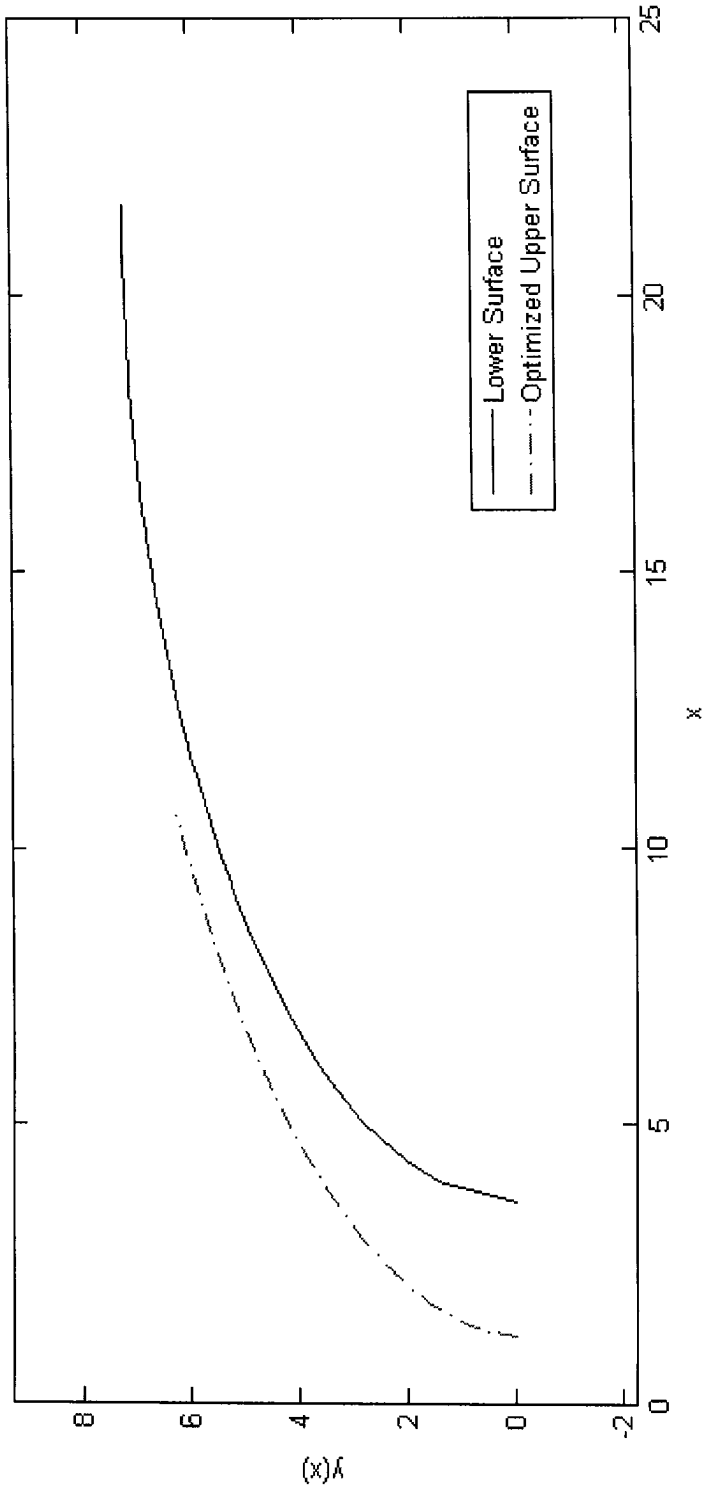

Lower Surface
Optimized Upper Surface
x
y(x)
25
20
15
10
5
0
-2
8
6
4
2
0

FIG. 14

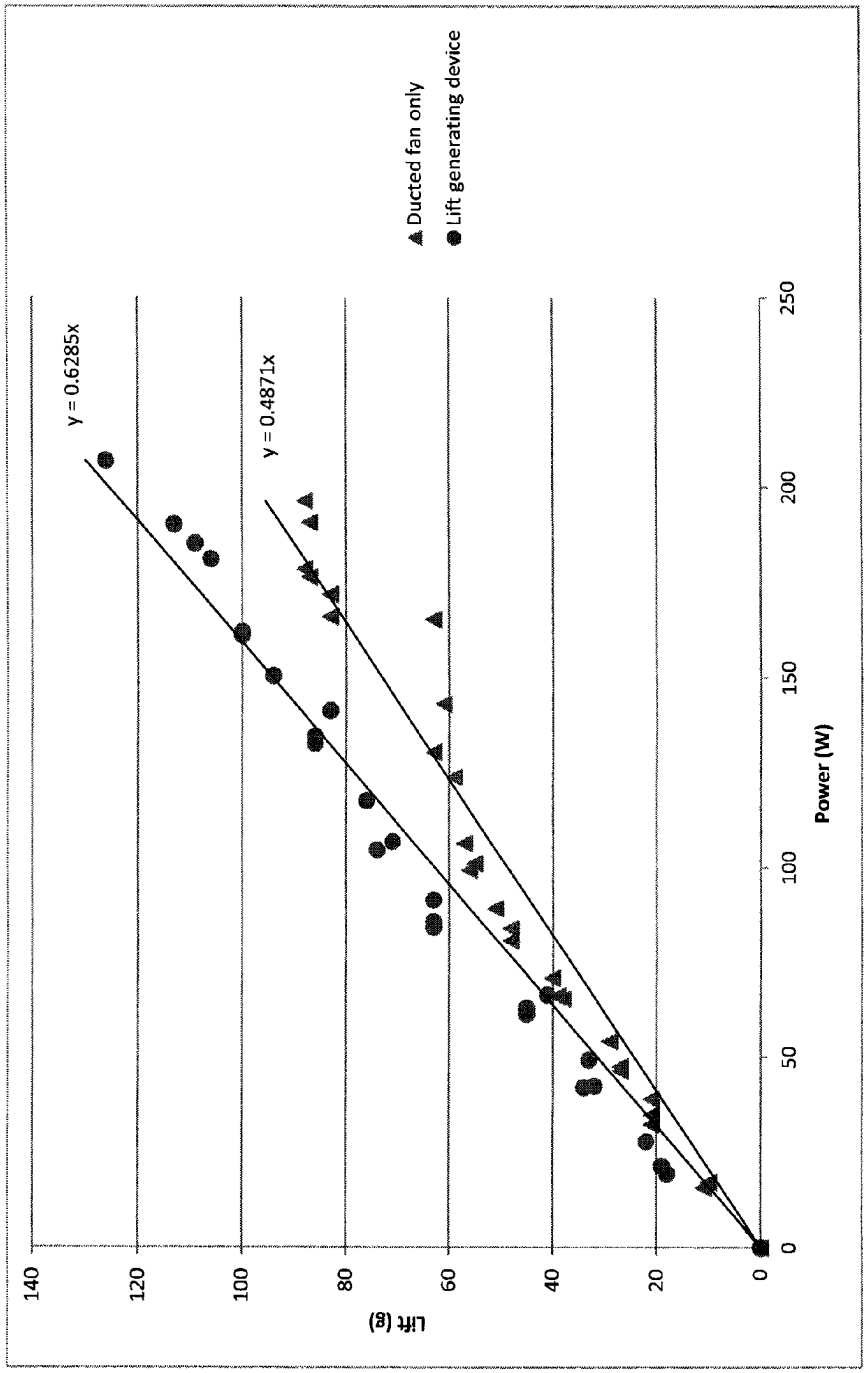

Ducted fan only
Lift generating device
y = 0.6285x
y = 0.4871x
Lift (g)
Power (W)
0
20
40
60
80
100
120
140
0
50
100
150
200
250

FIG. 15

LIFT GENERATING DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the benefit of U.S. Provisional Patent Application Ser. No. 61/534,530 filed on Sep. 14, 2011. The specification and drawings of the provisional patent application are specifically incorporated by reference herein.

TECHNICAL FIELD

Embodiments of the present invention generally relate to aircraft lift mechanisms and, more specifically, to lift generating devices for vertical flight vehicles.

BACKGROUND

When used on airplanes, propellers, ducted fans, and turbines provide the air speed required for the wings to generate lift. Current vertical flight vehicle configurations (e.g., VTOL) may propel the air downwards in order to generate lift. The speed of the airflow does not create any lift. The incoming air is directly suctioned from the surroundings by a propeller. Although aircraft that fly vertically by using various types of propelling devices may have great maneuverability and also the capacity for take-off and landing on relatively small areas, such aircraft consume high amounts of energy in order to generate the required lift, which makes them somewhat inefficient compared to other types of aircraft. The efficiency of the system is further limited by the air drag at high revolution rates.

SUMMARY

The disclosed embodiments are directed to a vertical flight vehicle configuration that includes a series of surfaces of revolution in the shape of funnels. The geometry of the system redirects the incoming airflow of a propelling device, creating extra lift during flight.

In exemplary embodiments, the efficiency of propelling devices used for vertical flight can be increased by redirecting the airflow through a series of surfaces of revolution. Redirecting the airflow creates a difference in pressure on the surface thereby generating a lifting force. This lifting force can be added to the force already produced by the propelling device resulting in a considerable increase in system efficiency.

In one embodiment, a lift generating device is provided for enabling vertical flight. The lift generating device includes a first revolution structure having a funnel shape and a second revolution structure having a funnel shape. Each revolution structure includes a circular open surface and an open stem. The second revolution structure is attached to the first revolution structure by a plurality of connectors so as to form a gap between the first revolution structure and the second revolution structure. A propeller is coupled to an open stem at a bottom end of the first revolution structure and a motor is operatively connected to the propeller to provide power for the propeller to suction air into the lift generating device.

In one embodiment, a lift generating structure is provided for enabling vehicle vertical flight. The lift generating structure includes a lift generating device having a funnel-shaped first revolution structure and a funnel-shaped second revolution structure. The first revolution structure is mounted in the second revolution structure and separated by a predetermined gap, such that airflow on an upper side of the first revolution structure decreases air pressure on the lift generating device while air pressure on a lower side of the first revolution structure remains unchanged resulting in a lifting force. The lift generating structure further includes a fuselage structure and a holding structure affixed to the fuselage structure for mounting of the lift generating device. A plurality of motor and propeller combinations generates horizontal motion of the vehicle. A plurality of side arms is attached to the fuselage structure for supporting each of the motor and propeller combinations.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other advantages and aspects of the embodiments of the disclosure will become apparent and more readily appreciated from the following detailed description of the embodiments taken in conjunction with the accompanying drawings, as follows.

FIG. **1** illustrates a perspective front view of a lift generating device in accordance with an exemplary embodiment.

FIG. **2** illustrates a cross-sectional view of the lift generating device in accordance with an exemplary embodiment.

FIG. **3** illustrates an exploded view of the lift generating device in accordance with an exemplary embodiment.

FIG. **4** illustrates an exemplary embodiment of a flying vehicle using a single lift generating device.

FIG. **5** illustrates an exploded view of the flying vehicle in the embodiment of FIG. **4**.

FIG. **6** illustrates an exemplary embodiment of a flying vehicle using two lift generating devices.

FIG. **7** illustrates an exploded view of the flying vehicle in the embodiment of FIG. **6**.

FIG. **8** illustrates an exemplary embodiment of a flying vehicle using four lift generating devices.

FIG. **9** illustrates an exploded view of the flying vehicle in the embodiment of FIG. **8**.

FIG. **10** illustrates an exemplary lift generating device configuration having a plurality of thin ribs attached to each revolution surface in accordance with an alternate embodiment.

FIG. **11** illustrates an exploded view of the lift generating device configuration of FIG. **10**.

FIG. **12** illustrates geometric parameters of the lift generating configuration in an exemplary embodiment.

FIG. **13** illustrates the area of revolution of the gap between surfaces is the difference of the outer area between two similar cones.

FIG. **14** illustrates a plot of the profiles for the upper and lower surfaces of revolution.

FIG. **15** illustrates lift versus power test results comparison between a ducted fan and the lift generating device.

DETAILED DESCRIPTION

The following detailed description is provided as an enabling teaching of embodiments of the invention including the best, currently known embodiment. Those skilled in the relevant art will recognize that many changes can be made to the embodiments described, while still obtaining the beneficial results. It will also be apparent that some of the desired benefits of the embodiments described can be obtained by selecting some of the features of the embodiments without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the embodiments described are possible and may even be desirable in certain circumstances. Thus, the following

description is provided as illustrative of the principles of the invention and not in limitation thereof, since the scope of the invention is defined by the claims.

The embodiments described herein can be applied to quadrotors and other vehicles that engage in vertical flight. Quadrotors are generally referred to as rotorcraft as opposed to fixed wing aircraft. The embodiments apply to quadrotors since such vehicles move by pitch and roll. Vertical take off and landing (VTOL) vehicles can use the embodiments described if the geometry of VTOL vehicles is adjusted properly. In some embodiments, the lift generating device can be added to each rotor to increase the lift considerably. In other embodiments, there could be two or more lift generating devices generating lift (while counteracting each other's torque to prevent the vehicle from spinning), and a propeller moving the vehicle horizontally.

Quadrotors have several advantages over VTOL aircraft and helicopters that make the lift generating device better-suited to quadrotors. One advantage is that quadrotors use fixed pitch rotors and variations in motor speed for vehicle control. VTOL aircraft require complex mechanical control linkages for rotor actuation. Another advantage is that using four rotors ensures that each rotor has a smaller diameter than an equivalent main rotor on a helicopter resulting in less kinetic energy storage during flight. The lift generating devices and rotors can also be enclosed in exemplary embodiments.

FIGS. **1-3** show a lift generating device **10** for vertical flight in accordance with an exemplary embodiment. The lift generating device **10** includes a first revolution structure **20**, a second revolution structure **30** attached to the first revolution structure **20**, a gap **12** formed between the first revolution structure **20** and the second revolution structure **30**, a propeller **40** inserted into the bottom end of the first revolution structure **20**, and a motor **50** operatively connected to the propeller **40**.

The lift generating device **10** can redirect the incoming airflow through the first revolution structure **20** and the second revolution structure **30** before reaching the propeller **40**. This action can create a difference in pressure that generates an extra upwards force that is added to the lift already produced by the propeller **40**, increasing the efficiency considerably.

The first revolution structure **20** can have a funnel shape including a circular open surface **22** and an open stem **24** having a radius "r", both resulting from a 360 degree rotation of curve **26**.

The second revolution structure **30** can have a funnel shape including an open surface **32** and an open stem **34**, both resulting from a 360 degree rotation of curve **36**.

The shape of curve **36** can be optimized based on the shape of curve **26** in order to maintain the velocity of the airflow constant through the space between the first revolution structure **20** and the second revolution structure **30**.

The open stem **34** of the second revolution structure **30** can be inserted into the open stem **24** of the first revolution structure **20**, forming a gap **12** between the two structures.

Connectors **60** can be placed between the first revolution structure **20** and the second revolution structure **30**. The connectors **60** can be evenly distributed around the perimeter of the second revolution structure **30** such that the connectors do not disturb the airflow.

The size of the second revolution structure **30** can depend on the gap **12**. The gap **12** can be more than 10% the size of the radius "r", as long as the velocity of the airflow remains constant through the gap **12**.

The first revolution structure **20**, the second revolution structure **30**, and the connectors **60** can be made of a sturdy and lightweight material, for example, carbon fiber or fiberglass. The first revolution structure **20**, the second revolution structure **30**, and the connectors **60** can be made by molding, or injection molding.

The propeller **40** can be a commercially available propeller or ducted fan. The propeller **40** can have an outer radius equal to the radius "r". The propeller **40** can be connected on the open stem **34** of the first revolution structure **20**. The propeller **40** can provide the required power to suction air into the lift generation device **10**.

Due to the shape of the first revolution structure **20** and the second revolution structure **30**, the air can be forced to flow through the gap **12**. As a result, the air flowing on top of the first revolution structure **20** can decrease the pressure, while the pressure below the first revolution structure **20** remains unchanged, therefore extra lift can be generated by just redirecting the incoming air.

Flaps (not shown) can be attached near the gap **12** where the air enters. Nozzles (not shown) can be added near an air output in order to control the lift generation device **10**.

A rear rotor can be added to stop the rotation of the vehicle (not shown) in view of the angular momentum. The rear rotor can be a commercially available rear rotor, for example, a rear rotor for a helicopter. The rear rotor can allow the user to control the vehicle.

In another embodiment, a plurality of lift generation devices **10** can be installed on the same vehicle in order to provide more lift. In this alternate embodiment, if the rotation of one propeller **40** is inverted with respect to another propeller **40**, then the rotation would be cancelled and no extra rear rotor may be needed.

In some embodiments, the first revolution structure **20** and the second revolution structure **30** can be made using a plurality of thin "ribs" pasted (evenly separated) to a circular and hollow base where the propeller **40** may be introduced. Then, the structures can be covered with paper or plastic in order to resemble the original shape.

The regulation of the power of the propeller **40** regulates the force produced by the lift generation devices **10**, since it depends on the airflow through the propeller **40**.

FIGS. **4-9** illustrate exemplary embodiments of flying vehicles using lift generating device assemblies in various configurations. FIG. **4** illustrates an exemplary embodiment of a flying vehicle **170** using a single lift generating device **10**. This embodiment includes two motors **110** with propellers for horizontal motion in forward and backward directions. Connectors **120** connect the motors **110** to fuselage structure **140**. Payload **100** can include, but is not limited to, power systems such as batteries and engines, control systems such as onboard computers and communications systems, cameras, cargo, passengers, weapons, etc. Back motor **150** provides vehicle counter-rotation.

FIG. **5** illustrates an exploded view of the flying vehicle **170** in the embodiment of FIG. **4**. The lift generating device **10** is mounted in holding structure **142**, affixed at a first end of fuselage structure **140**, and includes a horizontally-oriented annular structure **142** (e.g., circular opening) for mounting of the lift generating device **10**. Attached to the holding structure **142** is a plurality of side arms **148**. The motor/propeller assembly **110** is secured to side arms **148** by connectors **120**. The back motor **150** is mounted in the circular opening in vertically-oriented annular structure **144** at the other end of fuselage **140** where the motor **150** is secured in place. Platform **146** supports payload structure **100**. Also shown in FIG. **5** is propeller **152** for back motor **150**.

FIG. **6** illustrates an exemplary embodiment of a flying vehicle **180** using a plurality of lift generating devices **10**. This embodiment includes two lift generating devices **10** and two motor/propeller assemblies **110** mounted by side arms **134** to frame-shaped fuselage structure **130**. Connectors **120** secure the motors **110** to fuselage structure **130**. Payload **100** is supported by a platform that forms the bottom surface of fuselage structure **130**. Fuselage structure includes a landing gear structure **136** that is mounted or affixed to the fuselage structure **130**.

FIG. **7** illustrates an exploded view of the flying vehicle **180** in the embodiment of FIG. **6**. The lift generating devices **10** are mounted in holding structures **132** affixed at both a first end and a second end of horizontal bar structure **138**. Each holding structure **132** includes a horizontally-oriented annular structure (e.g., circular opening) at each end for mounting of each lift generating device **10**. Attached to the horizontal bar structure **138** is a frame-shaped fuselage structure **130**. A plurality of side arms **134** are attached to the fuselage structure **130** for securing motor/propeller assemblies **110**. The motor/propeller assemblies **110** are secured to side arms **134** by connectors **120**. No back motor is needed in this embodiment since the rotation of each motor/propeller assembly is inverted with respect to the other motor/propeller assembly. Fuselage structure **130** supports payload structure **100** on a lower platform surface.

FIG. **8** illustrates an exemplary embodiment of a flying vehicle **190** using a plurality of lift generating devices **10**. This embodiment includes four lift generating devices **10** and resembles the configuration and function of a quadrotor. Payload **100** is supported by fuselage structure **160**. Landing gear structure **166** is mounted or otherwise affixed to fuselage structure **160**.

FIG. **9** illustrates an exploded view of the flying vehicle **190** in the embodiment of FIG. **8**. The lift generating devices **10** are mounted in each of four holding structures **162** mounted to fuselage **160** by side arms **164**. Each holding structure **162** includes a horizontally-oriented annular structure (e.g., circular opening) for mounting of each lift generating device **10**. The holding structures **162** are mounted to frame-shaped fuselage structure **160** by side arms **164** affixed to a top portion of fuselage structure **160** and oriented in an angular direction away from the fuselage structure **160**. Fuselage structure **160** includes a lower surface platform that supports payload structure **100**. Landing gear structure **166** can be attached or otherwise mounted to the fuselage structure **160**.

FIG. **10** illustrates an exemplary lift generating device configuration having a plurality of thin ribs attached to each revolution surface in a light weight alternate embodiment. The lift generating device **10***b* includes lower revolution surface **20***b* that is held in place by a plurality of evenly distributed ribs **22***b*. Lower revolution surface **20***b* can be pasted or glued to the circular array of ribs **22***b*. Upper revolution structure **30***b* can be pasted or glued to a circular array of evenly distributed ribs **32***b*. Ribs **32***b* hold upper revolution structure **30***b* in place. Ribs **22***b*, **32***b* provide the required profile for both the upper and lower surfaces of revolution, so that the paper, fabric, or plastic can be glued or otherwise attached to the ribs. The upper and lower revolution structures form a gap **12***b* as shown. Motor **50***b* provides torque to lift generating device **10***b*. Connectors (not shown) are evenly distributed between the upper and lower revolution structures.

FIG. **11** illustrates an exploded view of the lift generating device **10***b* in the embodiment of FIG. **10**. Circular ring **70***b* holds ribs **22***b* in place. An incision **24***b* is made in each of the plurality of ribs **22***b*. A plurality of vertical incisions **72***b* is evenly distributed around circular ring **70***b*. The incisions **24***b* of ribs **22***b* can be inserted into the incisions of circular ring **70***b* in order to attach ribs **22***b* to circular ring **70***b*. Circular ring **80***b* holds ribs **32***b* in place. An incision **34***b* is made in each of the plurality of ribs **32***b*. A plurality of vertical incisions **82***b* is evenly distributed around circular ring **80***b*. Each rib **32***b* includes a bottom surface **36***b* that can be inserted into incisions **82***b* of circular ring **80***b*. Circular and symmetric part **90***b* holds the ribs **32***b* and the motor **50***b* in place. A plurality of vertical incisions **94***b* is evenly distributed around circular part **90***b*. The incisions **34***b* of ribs **32***b* can be inserted into the incisions of circular part **90***b* in order to attach ribs **32***b* to circular part **90***b*. Hollow base **92***b* holds motor **50***b* in place. Motor **50***b* provides torque to rotate propeller **40***b*. Motor **50***b* is mounted to shaft **52***b* that passes through circular part **90***b* and ring **80***b* and rotates propeller **40***b*. Propeller **40***b* is attached to shaft **52***b* below circular ring **80***b*.

Both the lower revolution structure **20***b* and upper revolution structure **30***b* of the lift generating device **10***b* could be made of a light and thin material such as fabric, paper or plastic. The ribs **22***b*, **32***b* could be made from, but are not limited to, wood, aluminum, plastic, carbon fiber, etc. Circular rings **70***b*, **80***b* could be made from, but are not limited to wood, aluminum, plastic, carbon fiber, etc. Circular and symmetric part **90***b* could be made from, but is not limited to wood, aluminum, plastic, carbon fiber, etc.

The following disclosure describes the mathematical analysis supporting the increased efficiency of propelling devices used for vertical flight by redirecting the airflow through a series of surfaces of revolution. The effective area of a propelling device can be described according to its circumference by its outer and inner radii (R_o and R_i, respectively), where the inner radius describes the size of the axel where the blades connect. Therefore, the effective area can be expressed as

$$A=(R_o^2-R_i^2)\pi. \tag{1}$$

In order to generate lift by redirecting the airflow of a propelling device, the system has to make the air flow at a desired speed on top of a nearly horizontal surface so that a nearly vertical force is generated due to the fact that the pressure on the top side of the surface decreases, while it remains unchanged at the bottom. FIG. **12** illustrates geometric parameters of the lift generating configuration in an exemplary embodiment.

The lower surface can have any type of curve profile as long as it allows the air to "bend" from nearly horizontal at the entrance to vertical at the propelling device's outlet. One particular curve that complies with such requirement is a quarter of an ellipse, and the equation describing such curve is

$$y(x) = b\cdot\sqrt{1-\left(\frac{x-r_o}{a}\right)^2}, \tag{2}$$

for the same geometric parameters shown in FIG. **12** and where x is the horizontal distance from the rotation axis.

In order to provide the lowest pressure on top of the surface, the air should flow as fast as possible. Using the continuity equation for incompressible flow,

$$A_1v_1=A_2v_2, \tag{3}$$

it becomes evident that in order to increase the velocity, the area must decrease. However, if the air speed is faster at the

inlet than at the outlet, a fraction of the consumed power will be lost when making the air flow through a smaller area. It seems that the best option would be to maintain the speed of the air constant at all times so that no power is lost. So, if the speed should remain constant, the effective area should remain constant as well,

$$A_1=A_2=A=(R_o^2-R_i^2)\pi. \tag{4}$$

Therefore, the profile of the upper surface depends on the profile of the lower surface: the closest distance between any point on the lower surface to a point on the upper surface represents the length of the gap between surfaces at that point, and the area of revolution of this gap around the central axis can be found as the difference in outer area of two similar cones,

$$A_{gap}=(xs-x_g s_g)\pi, \tag{5}$$

where x, x_g, s, and s_g are the radii and slopes of the two similar cones. FIG. **13** illustrates the area of revolution of the gap between surfaces is the difference of the outer area between two similar cones.

It is possible to observe from FIG. **13** that, since both cones are similar,

$$\cos\theta = \frac{x}{s} = \frac{x_g}{s_g}.$$

The area of the gap must remain constant,

$$A_{gap}=A,$$

$$xs-x_g s_g=R_0^2-R_i^2. \tag{6}$$

In addition, since the gap is the closest distance between the surfaces, its direction must be normal (perpendicular) to the lower surface, and its magnitude is given by the difference between s and s_g. In order to obtain the normal vector describing the position and magnitude of the gap, the tangential vector must be found first. Taking the first derivative of Eq. (2),

$$y'(x) = \frac{dy}{dx}, \tag{7}$$

$$y'(x) = -\frac{b}{a^2}(x-r_o)\left[1-\left(\frac{x-r_o}{a}\right)^2\right]^{-1/2}, \text{ or}$$

$$y'(x) = -\frac{b^2}{a^3}\left(\frac{x-r_o}{y}\right). \tag{8}$$

Therefore, the magnitude of the tangent vector can be written as,

$$|y'(x)|=\sqrt{1^2+y'^2},$$

and the unit tangent vector becomes,

$$v_t(x) = \left[\frac{1}{|y'|} \quad \frac{y'}{|y'|}\right].$$

The normal vector can be found from the derivative of the tangential vector,

$$v_t'(x) = \left[\left(\frac{1}{|y'|}\right)' \quad \left(\frac{y'}{|y'|}\right)'\right]. \tag{9}$$

The magnitude of the vector above is

$$|v_t'(x)| = \sqrt{\left[\left(\frac{1}{|y'|}\right)'\right]^2+\left[\left(\frac{y'}{|y'|}\right)'\right]^2}.$$

Thus, the normal unit vector is given by

$$v_n(x) = \frac{v_t'}{|v_t'|}. \tag{10}$$

Any point on the curve that describes the lower surface can be expressed as

$$P=[xy(x)],$$

and the normal unit vector at that point is given by

$$P_n = P - v_n(x), \tag{11}$$

$$= [\, x_n \quad y_n(x_n)\,].$$

Therefore, it is possible to find the slope of the gap

$$m = \frac{y(x)-y_n(x_n)}{x-x_n}.$$

The intersection with the central axis of rotation of the line with slope m that passes through point P is

$$q(x)=y(x)-mx,$$

This intersection point is given by

$$P_q=[0q(x)],$$

so that it becomes possible to find the distance from P_q to P(=s for the bigger cone)

$$s(x)=\sqrt{(q(x)-y(x))^2+(0-x)^2},$$

Rearranging equation (6) in order to find the dimensions of the smaller cone,

$$x_g s_g=xs+R_i^2-R_o^2.$$

Also, since ($x_g=s_g\cos\theta$) and ($\cos\theta=x/s$)

$$s_g^2\cos\theta = xs + R_i^2 - R_o^2, \tag{12}$$

$$s_g^2 = (xs + R_i^2 - R_o^2)\frac{1}{\cos\theta},$$

$$s_g = \sqrt{(xs + R_i^2 - R_o^2)\frac{s}{x}}.$$

Finally, the size of the gap is simply given by

$$g=s-s_g. \tag{13}$$

As a result, it is possible to find the coordinates of any point on the upper surface that is the closest to the lower surface

given that the area of the gap must remain constant. The coordinates can be found as follows:

$$x_g = s_g \cos\theta = s_g \frac{x}{s}, \quad (14)$$

$$y_g = q - \sqrt{s_g^2 - x_g^2} = q - s_g \sqrt{1 - \frac{x^2}{s^2}}. \quad (15)$$

These coordinates represent the points of the curve describing the optimized upper surface of revolution for a given value of x.

FIG. **14** shows the profiles for the lower surface and the optimized upper surface of revolution for the following parameters:

R_o=3.6, R_i=1.2, r_o=21.6, a=18, and b=7.2.

In addition, some preliminary testing has been done in order to compare the efficiency of a propelling device vs. the efficiency of the new lift generating configuration. FIG. **15** shows the lift generated vs. power consumption results for a small ducted fan (lower plot) and for a rough prototype (non-optimized) of the device using the same ducted fan (upper plot).

It can be observed that the new lift generating device is about 29% more efficient than the ducted fan alone (even for a non-optimized prototype). This result clearly illustrates the potential benefits of implementing this configuration into VTOL and similar aircraft.

The corresponding structures, materials, acts, and equivalents of all mans plus function elements in any claims below are intended to include any structure, material, or acts for performing the function in combination with other claim elements as specifically claimed.

Those skilled in the art will appreciate that many modifications to the exemplary embodiments are possible without departing from the scope of the present invention. In addition, it is possible to use some of the features of the embodiments disclosed without the corresponding use of the other features. Accordingly, the foregoing description of the exemplary embodiments is provided for the purpose of illustrating the principles of the invention, and not in limitation thereof, since the scope of the invention is defined solely by the appended claims.

What is claimed:

1. A lift generating device for enabling vertical fligh, comprising:

a first revolution structure having a funnel shape and including a circular open surface and an open stem;

a second revolution structure having a funnel shape and including an open surface and an open stem, the second revolution structure attached to the first revolution structure by a plurality of connectors and spaced therefrom so as to form a gap of a size sufficient to provide a selected air flow velocity between the first revolution structure and the second revolution structure;

a propeller coupled to an open stem at a bottom end of the first revolution structure, the propeller being disposed between the first revolution structure and the second revolution structure; and

a motor operatively connected to the propeller to provide power for the propeller to suction air into the lift generating device such that a pressure difference is created so as to generate an upward directed force.

2. A lift generating structure for enabling vehicle vertical flight comprising;

a lift generating device including a funnel-shaped first revolution structure and a funnel-shaped second revolution structure, the first revolution structure mounted in the second revolution structure and separated by a predetermined gap, such that airflow on an upper side of the first revolution structure decreases air pressure on the lift generating device while air pressure on a lower side of the first revolution structure remains unchanged resulting in a lifting force; a propeller disposed between the first revolution structure and the second revolution structure for creating the airflow;

a fuselage structure including a support bar having a rectangular cross section;

a holding structure affixed to the fuselage structure for mounting of the lift generating device;

a plurality of motor and propeller combinations for generating horizontal motion of the vehicle; and

a plurality of side arms attached to the fuselage structure for supporting each of the motor and propeller combinations.

3. The lift generating structure of claim **2** further comprising a back motor mounted to the fuselage structure for generating a counter-rotation force.

4. The lift generating structure of claim **2** wherein the support bar includes a first end section and a second end section.

5. The lift generating structure of claim **4** wherein the holding structure comprises a horizontally-oriented annular component affixed to the first end section of the fuselage structure for mounting the lift generating device.

6. The lift generating structure of claim **4** further comprising a vertically-oriented annular holding structure affixed to the second end section of the fuselage structure for mounting the back motor.

7. The lift generating structure of claim **2** further comprising a plurality of connectors for attaching each of the motor and propeller combinations to the side arms.

8. The lift generating structure of claim **2** further comprising a vertical plate mounted to a side of the holding structure and a platform affixed to the vertical plate at a bottom section for supporting a payload.

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